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Social learning behind closed doors: evidence from seniority voting in the US Supreme Court

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Abstract

We analyze the implications of seniority voting on Supreme Court decision-making. We provide reduced-form evidence consistent with justices learning from the votes of their senior colleagues as determined by the voting mechanism. To explain this evidence, we develop and estimate a model of sequential voting with social learning. In the model, justices weigh spatial payoffs against the appropriateness of their disposition and consider their preferences, private information and the choices of senior justices. We recover estimates of judicial preferences, expertise and learning and use them to assess the quality of Supreme Court dispositions relative to alternative voting mechanisms. We find that the quality of the Court's dispositions would have improved under an anti-seniority rule, in which junior justices are allowed to vote first.

Keywords Courts, Political economy, Structural estimation, Formal modelling, Sequential voting

Paper type Research paper

The relevance of ideology behind salient explanations of Supreme Court decision-making is widely accepted. Still, when tasked with resolving disputes, justices must assess how the facts of the case adhere to the body of the law, as given by precedent and legal philosophies (Epstein and Knight, 1997; Richards and Kritzer, 2002; Stephenson, 2009; Kastellec, 2010).

The interpretation that justices make of the information at their disposal alters their case dispositions on top of their ideological biases, ultimately shaping Supreme Court rulings. The positive political theory of courts provides a formalization of this notion, where judicial behavior is driven by the interaction between justices' preferences and the information they possess about the facts of the case (e.g. Kornhauser, 1992; Cameron *et al.*, 2000; Iaryczower and Shum, 2012).

Most empirical studies of judicial decision-making, however, use preference measures derived from conventional ideal point models (e.g. Martin and Quinn, 2002), which assume behavior is a direct reflection of justices' ideology. By leaving no room for the role of information, the assumptions behind these measures directly contradict many of the implications of strategic theories of decision-making in courts and abstract from the institutional environment under which judicial decisions are made.

As multiple formal and informal rules govern Supreme Court's decisions, the value of information behind justices' decisions is a direct function of these constraints. Of particular importance are the norms that regulate the exchange of information in the privacy of the conference room, where the bulk of justices' interactions take place. The most consequential of these norms is given by a decision-making rule under which justices express their views and vote on the merits of a case in order of seniority. The collective outcomes of this

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sequential process are reached via plurality rule in the form of the Court's preliminary disposition on the merits as well as the author of the Court's opinion, who is selected from the majority coalition at conference.

In this paper, we analyze the effect of the Supreme Court's seniority voting on justices' behavior in conference. On the one hand, the seniority voting mechanism provides junior justices the opportunity to learn from their senior colleagues relevant information about the facts of the case and how they align with the underlying body of the law. On the other, learning from seniors with different preferences and expertise might induce juniors to bias their choice toward a disposition at odds with the context of the law. Furthermore, the fixed nature of the voting order immediately prevents junior justices from influencing their seniors at conference. Some of the consequences of the seniority voting norm at conference have been well understood by justices in the past, as Rehnquist noted: *"… a junior justice's views were seldom commented upon because votes had already been cast up the line"* (Rehnquist, 2002 as cited in Segal and Spaeth, 2002). Stevens echoed Rehnquist's opinion in his memoirs, when contrasting seniority voting with an anti-seniority mechanism: *"… as the most junior justices, we thought our opportunity to persuade our more senior colleagues was lessened once they had announced their votes"* (Stevens, 2011).

Although what transpires at conference is not a matter of public record, we obtain conference voting records for the period 1946–1993 from the private papers that a few justices decide to make public after leaving the bench [1]. With this data in hand, we present reduced-form evidence of vote dependence consistent with learning as determined by the seniority voting order. To quantify the relative influence of learning in the Court we write down and estimate a model of sequential voting at conference that allows us to assess the amount of information extracted from the votes of senior justices and the impact of this information on the quality of the Court's dispositions.

We show that a significant portion of the observed dependence between senior and junior justices' votes can be attributed to seniority voting. Furthermore, we provide evidence that the effects of the seniority voting order on conference voting are not driven by a behavioral tendency to conform with the majority, which would inhibit meaningful information aggregation through learning.

To explain the previous evidence, we develop a sequential voting model that captures the main features of conference voting at the Supreme Court. In the model, justices care about choosing a disposition close to their ideal position, as in the standard spatial voting model, but also care about the appropriateness of their decision. Justices vote under incomplete information and incorporate their preferences, public and private information (as in Iaryczower and Shum, 2012), as well as the voting history of senior justices (as in López-Moctezuma, 2026). We structurally estimate the model parameters by fitting justices' behavior in the model to their conference voting records.

We recover the values of justices' preferences that account for justices' interdependence via the seniority voting rule. Similar to standard ideal point models, our estimates reflect the ideological biases and partisanship of justices in the data. In addition, they also account for non-ideological characteristics that affect justices' preferences over dispositions. We also provide estimates of justices' expertise that incorporate the exchange of information among justices throughout the voting sequence. These estimates imply that the quality of Supreme Court nominees has significantly increased over time, with junior justices having more precise information than their senior counterparts.

We use the estimated parameters of the model to provide a measure of social learning under seniority voting and assess its impact on the quality of the Court's decisions at conference [2]. We compare the quality of conference decisions under the observed seniority

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rule relative to two counterfactual voting mechanisms: *sincere* voting, in which justices follow their private information, ignoring the votes of others, and *anti-seniority* voting, in which justices vote sequentially, with junior justices voting first. We find both sequential mechanisms to be better at aggregating relevant information through justices' actions, as they outperform *sincere* voting. However, between sequential mechanisms, *anti-seniority* voting is significantly superior to the current seniority rule. These results are driven by the fact that junior justices possess better information than seniors, which is incorporated via social learning.

We contrast our *sequential* model with an alternative *conformity* model in which vote dependence reflects a desire to align with the conference majority, rather than learning from the information conveyed by senior justices' votes. We show that the *sequential* model with learning fits conference voting data significantly better than a framework in which justices care about casting a vote for the winning alternative.

1. Related literature

Our paper contributes to three distinct strands of literature. First, we build on the judicial politics literature that develops measures of justices' ideological preferences and qualifications (e.g. Segal and Cover, 1989; Segal *et al.*, 1995). In particular, Martin and Quinn (2002) recover time-varying estimates of justices' ideologies from individual votes inferred from case opinions, under the assumption that justices' decisions sincerely reflect their ideological preferences. Like this prior work, we recover Bayesian estimates of justices' preferences that evolve over their tenure. However, our approach explicitly incorporates the seniority-based voting mechanism, allowing us to capture both ideological and non-ideological considerations. In addition, we provide novel estimates of judicial expertise by disentangling the precision of justices' private information from the shared information revealed during the voting sequence. These estimates allow us to assess the informational environment faced by justices and their ability to process it.

By accounting for the joint influence of institutional decision rules and peer behavior, our analysis highlights justices as sophisticated decision-makers during conference proceedings. This view complements existing findings on judicial sophistication at other stages of the Supreme Court's decision-making process, such as case selection (*certiorari*) (e.g. Boucher and Segal, 1995; Caldeira *et al.*, 1999) and oral argument (Johnson *et al.*, 2006).

Second, we contribute to the literature on the structural estimation of committee decision-making. Our work builds on the framework developed by Iaryczower and Shum (2012), who estimate a simultaneous voting model of the Supreme Court using votes inferred from case opinions. We extend their approach by incorporating sequential voting, which accurately reflects the institutional norms of Supreme Court conferences. Similar to the deliberation framework of FOMC deliberations in López-Moctezuma (2026), we estimate a model of sequential decision-making but incorporate the majority rule aggregation used in conference voting.

We also relate to the work of Spenkuch *et al.* (2018), who model sequential roll-call voting in the U.S. House of Representatives. However, the mechanism driving the effects of sequential voting in our setting differs from theirs. While their model emphasizes strategic free-riding to avoid the party-line to vote sincerely, we focus on how sequential voting facilitates preference and information sharing among decision-makers.

Third, we introduce an empirical framework to quantify social learning with heterogeneous voters, as in the theoretical framework of Smith and Sørensen (2000). Our approach allows us to evaluate the extent of learning under a binding voting rule. In doing so, our model connects to the theoretical literature on social learning with externalities

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(e.g. Battaglini, 2005; Ali and Kartik, 2012), where agents weigh both private signals and the informational content of others' actions in strategic environments.

2. Conference voting and reduced-form evidence

2.1 Seniority voting at conference

Our analysis focuses on the seniority voting mechanism at conference after cases are selected from the petitions that arrive through *certiorari* and argued in open Court. These proceedings happen in private meetings with only the nine justices allowed to participate. Decision-making at conference represent justices' first action toward deciding the cases they hear (Dickson *et al.*, 2001). Moreover, as justices seem to follow closely the norm of not directly discussing cases prior to conference, conference votes captures a justice's initial leaning on a case (Black and Johnson, 2018). These initial positions are not only implemented behind closed doors but are preliminary, non-binding and open to change [3].

Decision-making on the merits typically starts with the Chief justice stating his opinion on the case and voting on the merits. Next, the associate justices give their opinion and vote in descending order of seniority. After every justice casts their vote, a majority is determined either in favor of affirming or reversing the lower court decision. The most senior justice in the majority assigns the authorship of the Court's opinion to a member of the majority coalition, including themselves [4].

As the seniority voting norm is an informal rule implemented behind closed doors, there has been some debate about its longevity. Perry (2009), based on interviews with four justices, argues that seniority voting has always been the norm at conference. Dickson *et al.* (2001) argues, however, that justices deliberated in order of seniority but voted in reverse order until 1946 and, for selected cases, as late as 1963. Consistent with the modal account, our main analysis assumes the seniority rule has been implemented for all cases in the post-WWII period [5].

We analyze conference votes for all formally decided cases from 1946 to 1993. Individual votes are coded based on the actual decision on the merits, i.e. whether justices voted to affirm or reverse the decision of the lower courts. This classification arises naturally from justices' choices at conference. Importantly, all our results are robust to the classification of these votes into "liberal" or "conservative," as manually classified in the Supreme Court Database (Spaeth *et al.*, 2018) (see Section E.1 in the Supplementary Online Appendix).

Conference votes for the period 1946–1985 are drawn from Spaeth's Vinson-Warren Supreme Court Judicial Database (Spaeth, 2002) and the Expanded Burger Court Judicial Database (Spaeth, 2007). For the period 1986–1993, we manually code conference votes using the docket sheets of Justice Blackmun, which have been digitized by Epstein *et al.* (2007) [6].

We exclude individual votes in which justices were absent during conference voting. These missing votes include a small number of instances, 1.3% for all justices and 2.6% for chief justices, in which a justice explicitly passed at conference and voted last. In Section I of the Supplementary Online Appendix, we add these cases to our analysis and quantify the informational gains associated with departures from the seniority voting protocol by more senior justices.

By the time justices sit together for a preliminary discussion and vote on the merits; justices already possess substantial information about the facts of the case as disseminated through the record of the lower Court proceedings, the case selection process and the oral arguments from the parties of the case. Our analysis of seniority voting at conference is conditional on the information already exchanged before the conference on the merits.

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Justices' private papers allow us to control for the information included in the case selection process that is exchanged among justices before a vote on the merits takes place. For this purpose, we include the individual voting behavior of justices on *certiorari* (i.e. case selection stage), which captures justices' individual disposition to review a case. We code justices' vote on *certiorari* as either grant (`cert_grant`), deny (`cert_deny`) or join-three (`cert_join`). A join-three vote captures the instances in which justices vote to grant review conditional on three other justices voting to grant [7]. To account for additional differences in agenda-setting powers in case selection, we also include Chief Justice dummies (`chief_FMVinson`, `chief_EWarren`, `chief_WEBurger`, `chief_WHRehnquist`).

The information on the facts of the case is obtained from the Supreme Court Database (Spaeth *et al.*, 2018). We distinguish cases about three issue areas: *Criminal* (including criminal procedure cases); *Basic Rights* (including civil rights, First Amendment, due process, privacy and judicial power cases); and *Economics* (including attorneys, unions and economic activity cases). For each case, we include the characteristics of the Petitioner and Respondent. We classify Petitioners/Respondents into Federal Government (`federal.plaint`, `federal.resp`), Local Government (`local.plaint`, `local.resp`) and private parties (`private.plaint`, `private.resp`). We code whether the position advocated by the Petitioner is classified as liberal or conservative (`liberal.plaint`). We classify the type of law under consideration for each case as constitutional (`law.const`), federal (`law.fed`) and other (`law.others`). We include the source of each case that is discussed at conference and distinguish between cases in which the lower Court is an appelate (`source.app`) or a state court (`source.state`). In terms of the lower court decision, we also include whether there were instances of dissent from one or more members of the lower Court (`lcDisagreement`). Table A.1 in the Supplementary Online Appendix presents a summary information of these case characteristics divided by Chief Justice.

Justices' characteristics that might affect their dispositions at conference come from the US Supreme Court justices Database (Epstein *et al.*, 2018). We include party affiliation (`dem.reg`), party of the nominating President (`dem.nom`), prior experience in academia (`prior.aca`), in government (`prior.gov`) and in the judiciary (`prior.jud`). Regarding education, we control whether a justice attended graduate school (`grad`). We also include the Segal-Cover scores that capture the ideology (`scideol`) and qualifications (`scqual`) of justices during their time as Supreme Court nominees. Our analysis accounts for the influence of justices' experience on the bench (`experience`), measured in years since confirmation. Table A.2 in the Supplementary Online Appendix presents summary statistics of these justice characteristics by Chief Justice.

2.2 Reduced-form evidence of learning

We present reduced-form evidence consistent with social learning in conference voting, conditional on both observed and unobserved characteristics of justices and cases. A key empirical implication of learning through seniority voting is serial dependence between senior and junior votes. To test this implication, we adapt the approach of Spenkuch *et al.* (2018), who examine for the presence of social learning in roll-call votes among US House representatives cast under a sequential alphabetical voting procedure.

We recover this dependence by estimating equations of the form:

$$v_{it} = \lambda_{i,c[t]} + \xi_{it} + \theta \overline{v}_{j<i,t} + \mathbf{X}_{it}\gamma + \epsilon_{it}, \tag{1}$$

where v_{it} takes the value of one if justice i votes to affirm case t, and takes the value of zero otherwise. $\overline{v}_{j<i,t}$ denotes the share of senior justices $j < i$ who voted to affirm. Equation (1)

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controls for justice/natural court fixed effects ($\lambda_{i,c[t]}$), justice i's voting-order effects (ξ_{it}) and case/justice observables ($\boldsymbol{X}_{it}$) [8]. By controlling for justice/natural court fixed effects, we isolate the serial dependence within justices for a natural court, which is defined as the set of cases for which the court composition remains fixed with no personnel changes [9]. Controlling for justices' position in the voting order accounts for the different number of votes each justice observes, and as such, incorporates the heterogeneity in the amount of information at their disposal.

We are interested in the parameter θ capturing the degree of serial dependence between senior and junior votes. However, given the observational nature of our data, this parameter could also capture the potential effect of unobservables that vary within justices across cases, which would not be absorbed by the fixed effects in equation (1). These unobservables could be present in the form of preference similarity between seniors and juniors or of information justices possess that is correlated across cases [10].

To disentangle the vote dependence that can be attributed to learning effects induced by the seniority voting order from other unobservables, we exploit justice-level variation in the voting order at conference.

First, we compare the estimates of θ from equation (1) to the implied degree of vote dependence that cannot be attributed to the seniority voting order. For this purpose, we construct a null distribution of vote dependence, where we randomly assign justices to different voting positions while keeping fixed their observed votes in the data, v_{it}. We then recompute $\overline{v}_{j<i,t}$ for $j < i$ in the generated voting order. We simulate this process 1, 000 times and recover a null distribution of θ that, by design, excludes the serial correlation in votes induced by seniority voting while capturing other sources of vote correlation present in the data. We compare this *null distribution* to the estimated serial correlation under the observed seniority voting procedure in the data [11]. Whenever the serial correlation observed under seniority voting is an outlier of the *null distribution*, it indicates a larger vote dependence driven by the seniority voting order compared to that from other factors.

We account for the potential correlation between justices' ideological affinities and their position in the seniority-based voting sequence, which can arise from the pattern of Supreme Court nominations across presidential administrations. In particular, justices who are close in seniority are more likely to have been nominated by the same president and may therefore be more ideologically aligned. To address this issue, we present results based on permutations of the voting order not only across all positions but also within groups of justices appointed by the same president, thereby preserving the ideological structure in the null distribution. Additionally, we conduct permutations within ideologically similar groups, defined by the political party of the nominating president and by Segal-Cover ideological scores [12].

Figure 1 presents the main results from this exercise [13]. The dashed line in the figure indicates an estimated increase of 0.225 points-or a 38% change-in the probability that a junior justice affirms the lower court decision, associated with a one-standard-deviation increase in the share of senior justices supporting affirmation (from 41% to 77%). To ease interpretation, we compute the marginal effect of a senior affirm vote on junior choices, which is given by $\frac{\theta}{n_{j<i,t}}$, where $n_{j<i,t}$ denote the number of senior justices who vote before justice i. As $n_{j<i,t}$ varies with the voting order, this marginal effect varies across justices. We present this effect by position in Figure A.1 in the Supplementary Online Appendix. On average, an additional affirm vote increases the probability that a junior justice votes to affirm in 0.19 points, which represent a 42% increase relative to the mean affirming probability in the data of 0.45 This effect is significantly larger (p-value < 0.01) under the observed seniority voting rule than under the *null distribution* generated by permuting the voting order across positions (*All*).

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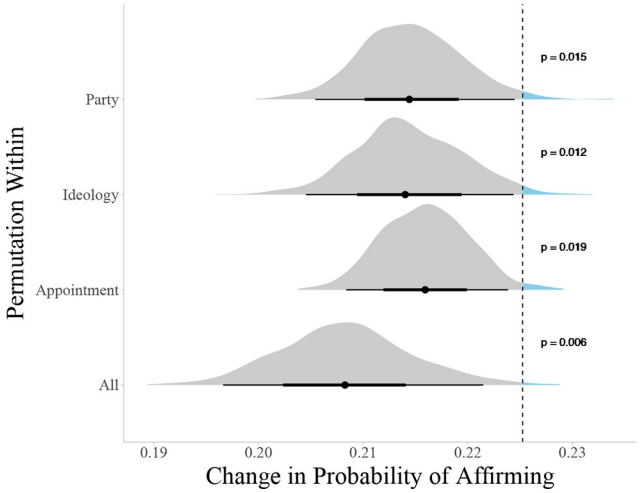


Figure 1. Effect of the seniority voting rule on learning

Note: The dashed lines represent the estimated change in the probability of affirming given equation (1), when we move the share of senior justices voting to affirm in one standard deviation from the mean (i.e. from 0.41 to 0.77) The density represents the null distribution of this effect under random voting order for 10, 000 simulations. In gray (blue), we show all the simulated values below (above) the estimated effect under seniority voting. Each density represents one type of permutation: *All* for permutation across all justices, *Appointment* for permutation within nominating president, *Party* for permutation within party at the time of nomination and *Ideology* for permutation within ideologically similar cluster (i.e. liberals and conservatives)

Moreover, the effect remains statistically significant even after we constrain permutations within more ideologically coherent subsets. Specifically, we find a p-value below 0.02 when permutations are restricted within: (*i*) justices nominated by the same president (*Appointment*), (*ii*) justices nominated by presidents of the same party (*Party*) and (*iii*) justices with similar Segal-Cover ideology scores (*Ideology*) [14].

In Supplementary Online Appendix B, we replicate the estimation of equation (1) and construct the corresponding *null distribution* using synthetic data generated under alternative behavioral models in which vote dependence is driven by sequential learning, ideological similarity or common information. We show that *(i)* the random-voting-order null distribution captures the serial correlation induced by ideological alignment and shared information, and *(ii)* estimated dependence that significantly exceeds this null distribution is consistent only with learning under seniority voting, rather than with these alternative mechanisms.

Supplementary Online Appendix A.3 provides additional evidence of vote dependence by examining how a justice's departure affects the behavior of junior colleagues. Unlike the main analysis, the before-and-after comparison around a senior justice's exit holds the seniority and the voting profile of remaining senior justices constant, while changing the voting position of junior justices (by one position), isolating the influence of the departing justice. We find significant effects across all junior justices, with larger impacts for those closest in seniority to the departing justice [15]. As in the main analysis, these effects exceed those generated under the null distribution based on random voting order permutations (Figure A.4).

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This reduced-form evidence suggests that justices incorporate the votes of their senior colleagues via the seniority voting order into their own decisions, beyond what would be expected from preference similarity and common information alone. Nonetheless, psychological accounts of conformity imply that social pressure may also drive vote dependence, regardless of the informational content of seniors' choices (Asch, 1956). Under majority rule, one might argue that such pressures apply naturally to majority voting decisions, as aligning with the majority means joining the winning side (Callander, 2008). Under this alternative interpretation, vote dependence may reflect a desire to align with the majority, independent of the outcome's ideological bias or quality, either from the enjoyment of supporting the winning alternative or because of extrinsic incentives such as avoiding the cost of writing dissenting opinions or gaining reputational benefits from presenting a unified front [16].

However, our reduced-form evidence suggests that seniority effects in Supreme Court conference voting are not simply the product of conformity pressures that impede information aggregation. This conclusion is reinforced in Supplementary Online Appendix H, where we show that a social-learning model fits the Court's voting patterns significantly better than a *conformity* model in which justices derive utility from aligning with the majority decision.

If justices were primarily motivated by a desire to join the winning coalition, they would value senior votes only for their contribution to the vote tally, i.e. their impact on the probability of supporting the eventual majority. In that case, the characteristics of senior justices would be irrelevant, conditional on their vote choice. Instead, as shown in Figure A.2 and Column (2) of Supplementary Online Appendix Table A.4, junior justices assign greater weight to the votes of ideologically similar seniors and those perceived as more qualified, suggesting that vote dependence reflects selective social learning rather than uniform conformity consistent with social pressures.

Moreover, when junior justices know their vote is not pivotal – that is, when a majority of senior justices has already supported one alternative – social pressure to conform should be strongest. Yet, we find no evidence that observing the winning alternative undermines information aggregation. Column (3) of Table A.4 shows that vote dependence remains similar even after excluding cases where junior justices are no longer pivotal. In this restricted sample, an additional affirm vote raises the probability of affirming by 0.15, compared to 0.19 in the full sample.

Second, in Supplementary Online Appendix I, we leverage the rare instances of passing at conference – when justices wait until all others have voted and can observe the vote tally and the majority outcome – to assess whether such behavior increases the likelihood of aligning with the winning alternative, as implied by conformity incentives [17]. However, Table 1.1 shows that the probability of voting with the majority is not significantly different under passing compared to the original seniority-based voting order.

Finally, under a pure conformity motive, we would expect a sharp increase in the likelihood that junior justices join the majority once their vote is no longer pivotal and the outcome is fully known. In this case, the difference between observing four versus five senior votes in favor of an alternative should primarily reflect conformity pressure, with the marginal informational value of the fifth vote playing a minimal role. To test this implication, we examine whether vote dependence increases discontinuously once a majority coalition has clearly formed, that is, when five senior justices support the same outcome. We find no such discontinuity, suggesting that conformity is not the primary driver of the observed voting patterns.

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To test for discontinuous changes in vote dependence once a majority coalition forms, we estimate the following regression:

$$v_{it} = \widetilde{\lambda}_{i,c[t]} + \widetilde{\xi}_{it} + \sum_{j=1}^{i-1} \widetilde{\alpha}_j \text{affirm}_{it,j} + \sum_{j=1}^{i-1} \widetilde{\beta}_j \text{reverse}_{it,j} + \mathbf{X}_{it}\widetilde{\gamma} + \widetilde{\epsilon}_{it}, \quad (2)$$

where $\text{affirm}_{it,j}$ ($\text{reverse}_{it,j}$) is an indicator equal to one if at least j senior justices have voted to affirm (reverse). As in equation (1), we include justice-by-natural-court fixed effects, voting-order effects and case-justice covariates.

Under a pure conformity mechanism, we would expect a discontinuous jump once a majority forms, with $\alpha_5 > \alpha_4$ and $\beta_5 < \beta_4$, as the junior's vote becomes non-pivotal and serves only to signal alignment with the majority.

Figure 2 presents the results of this exercise. The left panel shows the estimates of the coefficients $\widetilde{\alpha}_j$ and $\widetilde{\beta}_j$ while the right panel shows the estimated differences $\widetilde{\alpha}_j - \widetilde{\alpha}_{j-1}$ and $\widetilde{\beta}_j - \widetilde{\beta}_{j-1}$ for $j = 1, \ldots, 8$. We find no statistically significant change at the pivotal point (five vs four prior votes) in favor of either affirming or reversing [18]. In fact, the relationship between the coefficient estimates and the voting order is consistent with a learning mechanism with heterogeneous justices, where later votes are, on average, less informative, albeit with a non-monotonic decline [19].

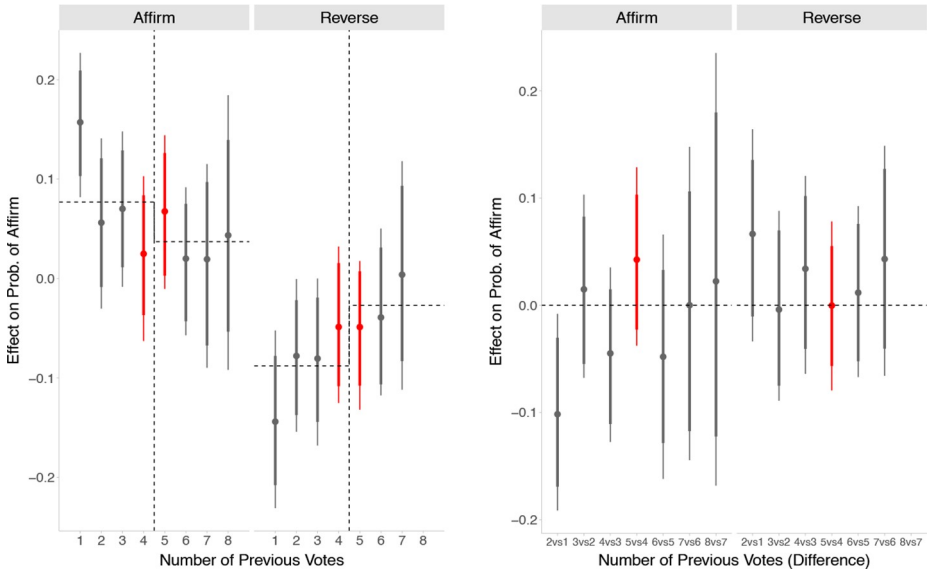


Figure 2. Conformity test (discontinuity at majority vote)

Note: Circles in the upper panel represent coefficient estimates of an affirm vote on indicator variables that equal one when there are at least $x = \{1, \ldots, 8\}$ prior votes in favor of either affirming or reversing, and zero otherwise. The regression controls for justice-by-natural-court fixed effects, voting order fixed effects, as well as justice and case characteristics. The lower panel presents the estimated differences between adjacent coefficients corresponding to the number of prior votes

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Building on the evidence of informational effects from seniority voting, we next quantify the extent of learning from senior votes relative to justices' private information and individual preferences. To do so, we estimate a structural model that links preferences, information (both common and private) and social learning to observed conference voting patterns. We leverage optimal behavior in a theoretical model to disentangle the sources of vote dependence under different counterfactual voting rules, which would not be feasible under a causal inference framework given the non-experimental nature of the data.

3. The model

We consider a court of N justices, $i = 1, \ldots, N$, who decide on the merits of each of $t = 1, \ldots, T$ independent cases that the Court chooses to review. For each case t, justice i decides between two alternatives, reversing (x_t^0) or affirming (x_t^1) the lower court decision, with $x_t^0, x_t^1 \in \mathbb{R}$. We denote $v_{it} = 0$ a vote in favor of x_t^0 and $v_{it} = 1$ a vote in favor of x_t^1. The conference outcome on the merits, v_t, is given by simple majority rule. Then, $v_t = 1$ if $\sum_i v_{it} \geq q \equiv \frac{N+1}{2}$ and 0, otherwise.

As in standard ideal point models, we assume that justices are *sincere*, as they care about the payoff from their own decisions [20]. Justice i has spatial preferences over alternatives relative to their ideal point $z_i \in \mathbb{R}$. In addition, they care about the appropriateness of their decision given the underlying body of the law. How the law should apply to each case t under consideration is represented by an underlying state variable $\omega_t \in \{0, 1\}$. For cases in which $\omega_t = 0$, the law is consistent with reversing the lower court decision, *de facto* favoring the Petitioner. For those cases in which $\omega_t = 1$, the correct application of the law is consistent with the lower Court's ruling, *de facto* favoring the Respondent. Given these preferences, the payoff of justice i from voting for x_t is:

$$U_{it}(x_t, \omega_{it}) = -\frac{1}{2}(z_i - x_t)^2 + \frac{1}{2} g_i(x_t, \omega_t), \tag{3}$$

where we assume quadratic policy preferences and an additional state-dependent payoff $g_i(x_t, \omega_t)$, such that $g_i(x_t, \omega_t) = \varsigma > 0$ whenever justice i chooses x_t^1 (x_t^0) if $\omega_t = 1$ ($\omega_t = 0$) and zero otherwise.

To mimic the voting mechanism at conference, we assume justices vote openly in a sequence with the Chief Justice voting in the first position, followed by the associate justices in order of seniority, with the most junior justice voting last. We rank members in this order so that justice i votes in the ith position.

For every case t under consideration, we assume that justice i lacks certainty about the correct application of the law, as reflected in the unobservability of ω_t. This uncertainty, even if residual, enables the justice to incorporate both public and private information, as well as the observed choices of other justices, into her own assessment. When justices do not share a common view about the correct legal outcome (i.e. when the non-ideological payoff of voting for each alternative is justice-specific) or place little weight on reaching the correct decision (i.e. $\varsigma \to 0$), their decisions are driven primarily by individual preferences over case alternatives. In this limiting case, the model converges to the standard *spatial voting* model used in ideal point estimation [21].

Before casting a vote, justice i receives three main pieces of information about how the law should be implemented. First, she considers publicly available information, including information from the case selection process leading up to the conference vote on the merits. This information is captured by the common prior belief that the law favors x_t^1 ($\omega_t = 1$), denoted by $\rho_t \in (0, 1)$.

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Second, justice i receives a private signal $s_{it}|\omega_t \sim \mathcal{N}(\omega_t, \sigma_{it}^2)$, with standard pdf and cdf, $\phi(\cdot)$ and $\Phi(\cdot)$, respectively. These signals are *i.i.d.* conditional on the state, with σ_{it} as a measure of justice i's ability/expertise to understand case t, as it captures the precision of her private information. For the vote on the merits, these signals can be thought of as the process by which a justice interprets, based on her ability and skills, how the evidence of the case relates to the true meaning of the law.

Third, given the seniority voting rule, justice i observes the voting history of her senior colleagues, starting with the Chief Justice. Let $\mathbf{h}_{it} \equiv (v_{1t}, \ldots, v_{(i-1)t}) \in \{0, 1\}^{i-1}$ denotes the voting history observed by justice i in case t, with $\mathbf{h}_{1t} \equiv \{\varnothing\}$.

3.1 Optimal voting strategies

Given the information available and expected payoffs, justice i votes to affirm the lower court decision ($v_{it} = 1$) whenever $E\left[U_{it}\left(x_t^1, \omega_{it}\right)|\Gamma_{it}\right] \geq E\left[U_{it}\left(x_t^0, \omega_{it}\right)|\Gamma_{it}\right]$, where Γ_{it} denotes the information justice i conditions her vote on. Thus, justice i prefers x_t^1 to x_t^0 if and only if:

$$Pr[\omega_t = 1|\Gamma_{it}] \geq \frac{1}{2} + \lambda_t(\kappa_t - z_i) \equiv 1 - \pi_{it}, \tag{4}$$

where $\kappa_t \equiv \frac{x_t^0 + x_t^1}{2}$ denotes the midpoint between alternatives in the policy space and $\lambda_t \equiv \frac{x_t^1 - x_t^0}{\varsigma}$ denotes the gap between affirming and reversing the lower court decision normalized by the payoff of voting for the appropriate disposition (ς).

The parameter π_{it} on the right-hand side of the optimal strategy (4) captures justice i's preference bias for case t in favor of either x_t^0 or x_t^1, which is a function of the location of case alternatives relative to the justice's ideal point in the policy space. As such, it varies by justice and case characteristics, biasing justice i's decision either toward affirming (i.e. $\pi_{it} \to 1$) or reversing (i.e. $\pi_{it} \to 0$) the lower court decision.

We do not impose justices' learning process in the model, and instead, estimate from the data the weight justices at conference put on the information contained in seniors' votes relative to their private information. We do this by allowing justices to be of two types depending on the information Γ_{it} they incorporate when voting: *sincere* ($t_i = S$) or *Bayesian* ($t_i = B$).

A *sincere* type casts a vote based only on her preference bias and private signal, $\Gamma_{it} = \{s_{it}\}$, disregarding the voting history. A *Bayesian* type conditions her vote on her preference bias, private signal, and on the voting history of senior justices, $\Gamma_{it} = \{s_{it}, h_{it}\}$.

For the *sincere* type, we can write her posterior on the state ω_t, $Pr[\omega_t = 1|s_{it}]$ via Bayes' rule as:

$$Pr[\omega_t = 1|s_{it}] = \frac{\rho_t \phi(s_{it}|\omega_t = 1)}{\rho_t \phi(s_{it}|\omega_t = 1) + (1 - \rho_t)\phi(s_{it}|\omega_t = 0)}$$

Given the normality assumption on the signal distribution, the optimal strategy of a *sincere* type is given by a unique cutpoint, which implies choosing $v_{it} = 1$ whenever:

$$s_{it} \geq \frac{1}{2} + \sigma_{it}^2 \left[\log\left(\frac{1 - \pi_{it}}{\pi_{it}}\right) + \log\left(\frac{1 - \rho_t}{\rho_t}\right)\right] \equiv s_{it}^{sin} \tag{5}$$

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A *Bayesian* type conditions her vote on their private signal and the voting history, $\Gamma_{it} = \{s_{it}, h_{it}\}$. Let $\gamma_{i,\omega_t} \equiv Pr[v_{it} = 1|\omega_t]$ denote the probability that justice i votes to affirm the lower court decision given the unobserved state ω_t. For this type of justice we can write $Pr[\omega_t = 1|\{s_{it}, h_{it}\}]$ via Bayes' rule as:

$$Pr[\omega_t = 1|\{s_{it}, h_{it}\}] = \frac{\rho_t \phi(s_{it}|\omega_t = 1)\prod_{j=1}^{i-1} \gamma_{j,1}^{v_{jt}}\left(1-\gamma_{j,1}\right)^{1-v_{jt}}}{\rho_t \phi(s_{it}|\omega_t = 1)\prod_{j=1}^{i-1} \gamma_{j,1}^{v_{jt}}\left(1-\gamma_{j,1}\right)^{1-v_{jt}} + (1-\rho_t)\phi(s_{it}|\omega_t = 0)\prod_{j=1}^{i-1} \gamma_{j,0}^{v_{jt}}\left(1-\gamma_{j,0}\right)^{1-v_{jt}}}$$

Then, their optimal decision is $v_{it} = 1$ whenever:

$$s_{it} \geq \frac{1}{2} + \sigma_{it}^2\left[\log\left(\frac{1-\pi_{it}}{\pi_{it}}\right) + \log\left(\frac{1-\rho_t}{\rho_t}\right) + \sum_{j=1}^{i-1}\log\Psi(\boldsymbol{h}_{jt})\right] \equiv s_{it}^{bayes} \tag{6}$$

where:

$$\Psi(\boldsymbol{h}_{jt}) \equiv \left[\frac{\gamma_{jt,0}}{\gamma_{jt,1}}\right]^{v_{jt}}\left[\frac{1-\gamma_{jt,0}}{1-\gamma_{jt,1}}\right]^{1-v_{jt}}, \tag{7}$$

captures the value of senior justice $j < i$ on justice i's decision. This value is determined by the relative likelihood with which justice j's vote matches the state ω_t, which in turn is a function of justice j's preference bias and expertise. In particular, justice j is more influential on junior justices' decisions if they possess a higher level of expertise ($\sigma_{jt} \to 0$) and more moderate preferences ($\pi_{jt} \to \frac{1}{2}$). Under these circumstances, justice j's decision has informational value because they make the appropriate decision ($v_{it} = \omega_t$) with high probability.

We assume that justice i's type is her private information. We also assume types are *i.i.d.* across both justices and cases. The probability that justice i is *Bayesian* is given by $\iota_i \equiv Pr[t_i = B]$. Thus, in the sequential model, justice i's probability of voting in favor of affirming in state ω_t, $\widetilde{\gamma}_{it,\omega_t}$ is given by a mixture distribution over types:

$$\widetilde{\gamma}_{it,\omega_t} = \iota_i \gamma_{it,\omega_t}\left(s_{it}^{bayes}\right) + (1-\iota_i)\gamma_{it,\omega_t}\left(s_{it}^{sin}\right), \tag{8}$$

where $\gamma_{it,\omega_t}\left(s_{it}^{bayes}\right) \equiv 1 - \Phi\left(\frac{s_{it}^{bayes} - \omega_t}{\sigma_{it}}\right)$ and $\gamma_{it,\omega_t}\left(s_{it}^{sin}\right) \equiv 1 - \Phi\left(\frac{s_{it}^{sin} - \omega_t}{\sigma_{it}}\right)$.

Given justices' optimal behavior, s_{it}^{sin} and s_{it}^{bayes} for the *sincere* and *Bayesian* types, respectively, we can write the likelihood of justices' sequence of votes in case t, v_t as:

$$Pr[v_t] = \sum_{\omega_t \in \{0,1\}} \rho_t^{\omega_t}(1-\rho_t)^{1-\omega_t}\prod_{i=1}^{N}\widetilde{\gamma}_{it,\omega_t}^{v_{it}}\left(1-\widetilde{\gamma}_{it,\omega_t}\right)^{1-v_{it}}. \tag{9}$$

4. Identification and estimation of model parameters

4.1 Intuition for identification

We provide a formal argument for the identification of all parameters in the *sequential voting* model in Section C.1 of the Supplementary Online Appendix. Additionally, we evaluate

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identification through a series of Monte Carlo simulations in Supplementary Online Appendix C.2. For various configurations of parameter values determined by justices' seniority, we simulate individual conference votes across independent cases and verify that the estimated parameters converge to their true values. Below, we offer the intuition behind how model parameters are identified from observed data.

The prior ρ_t is identified from the cases in which a majority of justices affirm the lower court decision. This identification is a feature of the information structure assumed in the model, where a high (low) value of ρ_t generates larger (smaller) private signals across all justices, consistent with $\omega_t = 1$ ($\omega_t = 0$). Similarly, the identification of a justice's expertise σ_{it} comes from the assumption of the signal distribution. As signals are unconditionally correlated across justices (since their mean is ω_t), higher priors will induce higher signals for all justices. Thus, a justice who systematically sides with the majority across cases will have a larger precision in the empirical model (i.e. lower σ_{it}).

The bias parameter π_{it} is identified from the variation in the proportion of affirm/reverse votes for any given justice across cases with different priors. In turn, justices who show low variability in the proportion of affirm/reverse votes are relatively more biased in the empirical model.

Finally, the relevant proportion of *Bayesian* types, ι_i, is identified from the degree of serial dependence on individual votes between senior and junior justices across cases, as computed in the reduced-form evidence. A higher dependence on senior justices' votes is rationalized in the model with a higher probability that a junior justice is *Bayesian*. Since both an increase in expertise (i.e. lower σ_{it}) and in the share of *Bayesian* types (i.e. higher ι_i) will generate more agreement in votes across justices; the identification of the type probability ι_i comes from the variation in voting histories observed by justices throughout the seniority sequence in non-unanimous cases. The differences in how justices respond to an observed vote history are independent of justices' quality (σ_{it}), but they are crucial to disentangle a *Bayesian* from a *sincere* type in the data.

4.2 Model estimation

We structurally estimate the model introduced in Section 3. Let $\mu_t \equiv \{\{\pi_{it}, \sigma_{it}, \iota_i\}_{i=1}^{N}, \rho_t\}$ be the vector of structural parameters for case t, capturing both justices' preferences, as well as the quality of both public and private information [22].

We estimate these parameters within a Bayesian framework, which allows us to recover the posterior distribution of $\boldsymbol{\mu}_t$ for each case $t = 1, \ldots, T$, using the data likelihood in equation (9) and a prior distribution over $\boldsymbol{\mu}_t$. The estimation algorithm involves two main interrelated steps. In an *inner loop*, we fix a trial value of the parameters $\boldsymbol{\mu}_t = \boldsymbol{\mu}_{0,t}$ and compute justices' optimal behavior, summarized by the cutpoints s_{it}^{exp} and s_{it}^{bayes}, which determine the votes' likelihood.

In an *outer loop*, we approximate the posterior distribution of the structural parameters that best fit the data using the Hamiltonian Monte Carlo method. We allow these structural parameters to flexibly depend on justice- and case-level covariates, imposing functional forms that respect the support of each parameter in the model. Supplementary Online Appendix D provides additional details on the estimation procedure.

5. Estimation results

We estimate the model separately for each of the main issues under consideration by the Court in the period under study. We analyze cases related to *Basic Rights*, *Economics* and

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Criminal issues. These issues cover around 95% of all observations in our sample of conference votes.

We focus the description of our results on the sample of *Basic Rights* cases, as they represent the modal issue in our data. Nevertheless, we highlight relevant findings across all issues. Section F of the Supplementary Online Appendix presents all the results for *Criminal* and Economic issues.

5.1 Data fit and model comparison

Before presenting the parameter estimates of the *sequential voting* model, we assess its predictive performance on conference voting data, both in- and out-of-sample. We evaluate the model's goodness of fit relative to three alternatives: (*i*) a *sincere* behavioral model that assumes votes are determined solely by justices' individual biases and private information ($\iota_i = 0$); (*ii*) a static (Clinton *et al.*, 2004) and (*iii*) a dynamic (Martin and Quinn, 2002) version of a spatial voting model, which assumes justices' conference votes are a reflection of justices' ideal points relative to the ideological location of the case alternatives.

The comparison with the *sincere* model highlights the added value of incorporating other justices' votes − through the seniority voting mechanism − into a framework where information interacts with individual preferences. In contrast, the comparison with the spatial voting models underscores the correlation in justices' behavior that arises from the informational content embedded in justices' votes beyond what is explained by ideological positioning alone. Estimation details for the three alternative models used for comparison are provided in Supplementary Online Appendix E.1.

The left panel of Figure 3 compares in-sample fit across models using the fraction of correctly classified recommendations as the evaluation metric [23]. The right panel fully accounts for model complexity by evaluating predictive performance on out-of-sample data using leave-one-out cross-validation (LOO). The measure of predictive accuracy for new data points is the expected log pointwise predictive density:

$$elpd_{loo} = \sum_{t=1}^{T} \sum_{i=1}^{N} \log Pr\left[\mathbf{v}_t | \mathbf{v}_t^{-i}\right], \tag{10}$$

where $Pr\left[\mathbf{v}_t | \mathbf{v}_t^{-i}\right] = \int Pr(\mathbf{v}_t | \boldsymbol{\mu}_t) Pr\left(\boldsymbol{\mu}_t | \mathbf{v}_t^{-i}\right) d\boldsymbol{\mu}_t$ is the posterior predictive distribution of conference votes for a given case t, $\mathbf{v}_t$, without the ith vote, v_{it} [24].

The *sequential voting* model achieves substantially better fit both in- and out-of-sample compared to the alternative specifications. By incorporating previously cast votes, it correctly predicts approximately 75% of individual policy recommendations, whereas the alternative models predict around 66%. Moreover, the sequential model yields a significantly higher out-of-sample predictive likelihood than both the *sincere* and *spatial voting* models, which perform similarly in terms of predictive accuracy.

The superior predictive performance of the *sequential voting* model stems from its ability to nest the core features of the alternative frameworks. It captures preference heterogeneity, as in *spatial voting* models, and accounts for variation in information quality, a central element of the *sincere* model. Crucially, it also leverages the informational content embedded in the votes of other justices, significantly improving model fit and underscoring the explanatory power of social learning in judicial decision-making.

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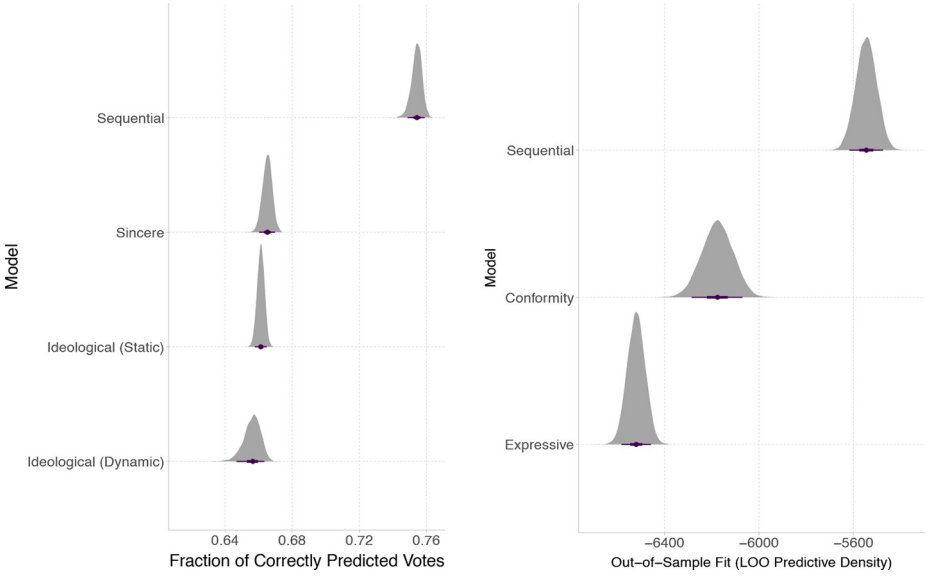


Figure 3. In-sample and out-of-sample fit across behavioral models

Note: The left panel shows in gray the posterior distribution of the fraction of correctly predicted recommendations under the different behavioral models. Solid circles depict posterior median and thick (thin) lines represent 50% (95%) credible intervals. The right panel shows posterior estimates of out of sample fit given by leave-one-out (LOO) predictive density given the data without the ith data point

5.2 Parameter estimates

We present our point estimates in the form of posterior medians and their uncertainty as posterior credible intervals. We focus the description of our results on individual-specific parameters for preference and expertise averaging out variation across cases. Figure F.1 in the Supplementary Online Appendix presents the evolution across terms of the common prior belief on how the law should be implemented in each case under consideration before any information is exchanged at conference (ρ_t).

5.2.1 Preference estimates. Our estimation strategy allows us to recover justices' underlying preference biases π_{it} as a function of both case-specific and individual-level characteristics [25]. Given the observed covariates, we can compute each justice's preference bias for every case in the data.

To separately disentangle the effect of justices' ideological preferences from case characteristics, we isolate the variation in π_{it} attributable to justices' ideal points, z_i, by focusing on the subset of cases in which the Petitioner's position is classified as liberal (Spaeth *et al.*, 2018). In Supplementary Online Appendix E.2, we show that, conditional on estimates of preference biases π_{it}, fixing the ideological direction of cases suffices to identify and recover z_i from the distribution of π_{it}. Under this restriction, larger (smaller) values of z_i denote more conservative (liberal) justices [26].

Figure 4 presents the posterior distribution of justices' ideal points, estimated using the subset of cases in which the Petitioner is classified as liberal. The relative ranking of justices along the ideological spectrum aligns closely with both anecdotal and systematic evidence on the political leanings of Supreme Court justices [27]. Figure 5 shows the correlation

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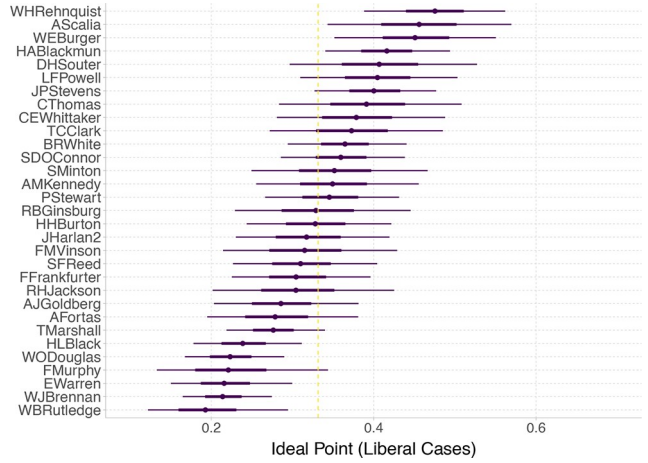


Figure 4. Ideal points z_i based on $\overline{\pi}_i \equiv \frac{1}{T}\sum t = 1^T \pi_{it}$ for liberal cases (basic rights)
Note(s): Each point in the right- (left-) hand panel depicts the posterior median of $\hat{\pi}_i$ after averaging over all (liberal) cases. Solid thick (thin) lines represent 50% (90%) credible intervals

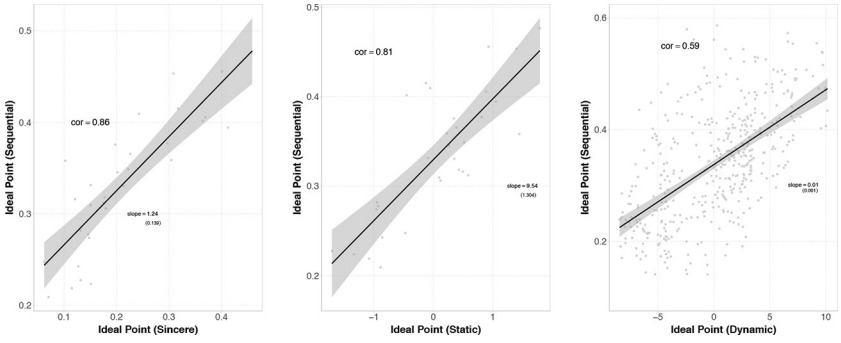


Figure 5. Correlation of preference parameters across models
Note: Each scatter plot correlates the posterior median of ideal points from the *sequential voting* model with the posterior median of the ideal point from the alternative model. The comparison with the *sincere* and static *spatial* voting models is done at the justice level. The comparison with the dynamic *spatial voting* is done at the justice/court term level. Solid lines represent linear fit with 95% confidence bands

between ideal point estimates from the *sequential voting* model and those obtained from the *sincere* and *spatial voting* models. We find a strong correlation between our estimates and the preference parameters from alternative discrete choice models. These results suggest that the *sequential voting* model effectively captures the key variation in behavior driven by justices' ideological positions captured with alternative frameworks.

5.2.2 Expertise estimates. Beyond ideological bias, individual voting behavior in our model also depends on each justice's ability to map the facts of a case onto the underlying body of law. As with preference estimates, the expertise measures we recover discount the

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influence of social learning [28]. We find that, once the sequential voting process is explicitly accounted for, junior justices possess more precise private information than their senior counterparts. Examining behavior across cases, our estimates suggest that the relative informational quality of more recent Supreme Court justices is higher. This pattern is illustrated in the left panel of Figure 6, where early nominees exhibit greater variance in their private signals (i.e. lower precision) compared to more recent appointees.

The right-hand panel of Figure 6 displays the average estimated ranking of justices along the expertise dimension. We find a consistent ordering of justices' expertise across issue areas, with Justices Burger, Scalia, Souter and Rehnquist emerging as the most accurate in terms of the quality of their private information. At the other end of the spectrum, Justices Black, Rutledge, Douglas and Goldberg rank lowest in informational precision [29].

5.3 The extent of social learning in the court

When deciding on their optimal strategies, justices who behave as *Bayesian* types consider not only their prior leanings and private information but also the informational content embedded in the votes of senior justices. As a result, the weight that junior justices place on the actions of their senior colleagues is reflected in the probability that a justice is drawn from the pool of Bayesian types, ι_i. Figure F.2 in the Supplementary Online Appendix presents the estimates of ι_i by justice across issue areas. Consistent with the reduced-form evidence and the superior predictive performance of the *sequential voting* model, the data rejects the hypothesis that voting behavior at conference is driven solely by preferences biases and private information. We estimate that the average probability a justice incorporates seniors' votes as a Bayesian is $\hat{\iota}_i = 0.4$, with similar estimates across issues. Furthermore, the lower bounds of the 90% credible intervals for all three issue areas exceed 0.3, and the entire posterior distribution lies above 0.2.

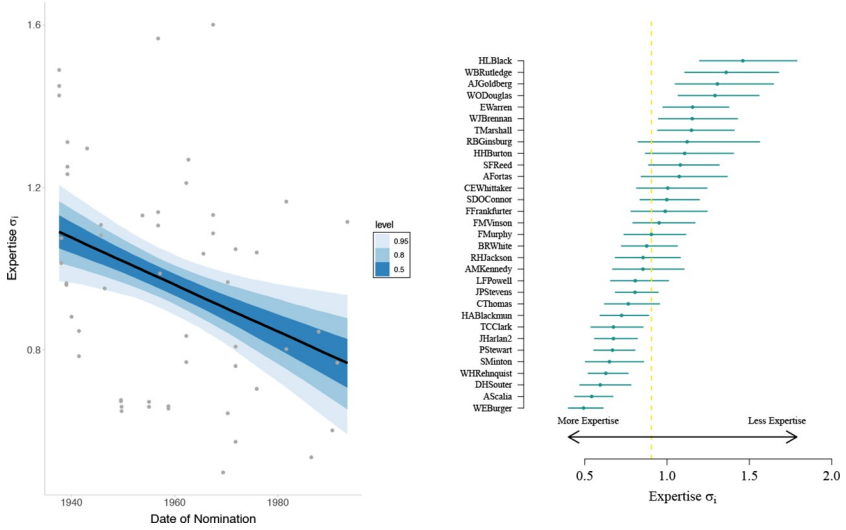


Figure 6. Estimates of expertise by date of nomination and justice

Note: Each point depicts the posterior median of $\hat{\sigma}_i$ after averaging over cases in the sample. Lines on the right-hand plot represent 90% credible intervals

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We combine our estimates of justices' biases, expertise and type probabilities to quantify the extent of social learning in the Court in a manner that is comparable across justices and cases. To this end, we compute the difference in voting behavior under the observed seniority-based sequential voting mechanism relative to a counterfactual scenario in which justices vote *sincerely*, that is, without incorporating the votes of their senior colleagues. Specifically, for each case in our sample:

- We draw the unobserved state ω_t conditional on ρ_t.
- We draw a signal for each justice from $\mathcal{N} \sim \left(\omega_t, \sigma_i^2\right)$.
- We compute justices' vote in conference under seniority voting, in which justices vote conditional on the voting history with probability ι_i (v_{it}^b).
- We compute a counterfactual scenario where justices vote *sincerely* (v_{it}^{sin}) with probability one.
- We compute the extent of social learning by justice as the probability that a vote under *sincere* behavior differs from that under learning from senior votes:

$$l_i = Pr\left(v_{it}^b \neq v_{it}^{sin}\right).$$

- We compute the extent of social learning in the Court as the probability that the Court's disposition (i.e. the majority vote at conference) differs when justices move from voting *sincerely* to learning from the voting history:

$$l_c = Pr\left(v_t^b \neq v_t^{sin}\right).$$

Figure 7 presents the extent of social learning in the Supreme Court for *Basic Rights* cases. The left panel displays the magnitude of social learning by justice, as well as for the Court's disposition (i.e. the conference majority), while the right panel summarizes learning for the average justice and for the Court's disposition across Supreme Court terms. On average, we find that the probability the Court's disposition changes between a scenario in which all justices vote *sincerely*, ignoring senior votes, and one in which they incorporate the observed voting history is 0.18.

Beyond this average effect, we uncover substantial heterogeneity in the extent of social learning, both across justices and over time. At one end of the spectrum, Justices Douglas, Murphy and Rehnquist appear least influenced by their senior colleagues, with switching probabilities below 0.08. At the other extreme, Justices Ginsburg, Kennedy and Whittaker are the most responsive to prior votes, with switching probabilities exceeding 0.34. [30]

The estimates for Chief Justices Vinson, Warren and Burger cannot be identified from the data, as they consistently voted first under the seniority voting protocol during their tenure. By contrast, estimates for Chief Justice Rehnquist are identified from his earlier tenure as an Associate Justice. In Supplementary Online Appendix I we show the robustness of our analysis to including the rare instances of passing in conference, where a justice waits to cast a vote until all other justices have voted (Johnson *et al.*, 2005). For example, Chief Justice Burger passed in approximately 6% of cases, and we estimate that he would have changed his vote relative to *sincere* voting in 19% of those instances, comparable to the average level of social learning observed among Associate Justices.

We find a significant shift in the extent of social learning in the Court over time, coinciding with the transition from Chief Justice Warren to Chief Justice Burger. During the

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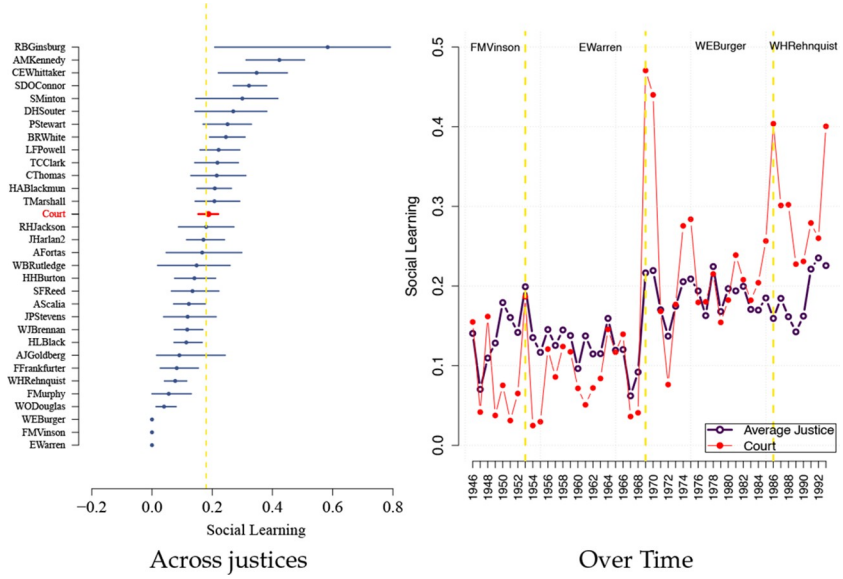


Figure 7. Social learning in the court

Note: Each point in the left-hand panel depicts the posterior median of $\hat{l}_i$ and lines represent 90% credible intervals. The right-hand panel plots the posterior median for $\hat{l}_c$ for the average justice and the majority in each natural Court

Vinson and Warren Courts, the probability of switching due to social learning is estimated at 0.13 for the average justice and 0.08 for the Court's disposition. In contrast, during the Burger and Rehnquist Courts, social learning reaches 0.19 for the average justice and 0.25 for the Court's disposition.

5.4 Quality of information aggregation

The voting outcome at conference determines both the winning coalition on the merits and the author of the Court's opinion. Given its importance and the extent of social learning, we assess the effectiveness of seniority voting in aggregating preferences and information by comparing it to alternative voting mechanisms.

We define the quality of a voting rule as the probability that the Court's decision aligns with the correct disposition of the case. Higher-quality rules thus minimize the probability of mistakes – i.e. instances in which the conference disposition v_t diverges from the true state of the law, ω_t. Although ω_t is unobserved, we can compute it as the sum of the probabilities that the Court reverses when it should affirm (with probability ρ_t) and that it affirms when it should reverse (with probability $1-\rho_t$). Formally:

$$Pr[v_t \neq \omega_t] = \rho_t Pr\left[\sum_{j=1}^{N} v_{jt} < q | \omega_t = 1\right] + (1-\rho_t) Pr\left[\sum_{j=1}^{N} v_{jt} \geq q | \omega_t = 0\right]. \quad (11)$$

Using our structural estimates, we compare the probability of mistakes under the observed seniority-based sequential voting to two alternatives: *sincere* voting and an *anti-seniority*

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rule. The comparison with *sincere* voting captures the informational gains from sequential voting. The *anti-seniority* comparison fixes sequential voting but addresses the consequences of having a voting order with junior justices leading the vote, a norm that, according to Dickson *et al.* (2001) appeared to be common before the Second World War.

The left panel of Figure 8 reports the average probability of Court mistakes under the three voting rules considered. On average, both sequential voting mechanisms yield lower error rates than *sincere* voting, underscoring the informational gains from incorporating the votes of other justices. However, among the sequential rules, the probability of mistakes is 43% lower under the *anti-seniority* rule than under the observed seniority rule – dropping from 0.14 to 0.08. This finding highlights the critical effect of the voting order on the effectiveness of information aggregation under sequential voting.

As with social learning, we find substantial variation over time in the relative quality of the Court's dispositions, as shown in the right panel of Figure 8. During the Vinson and Warren Courts, when social learning under seniority voting was weakest, the quality advantage of seniority voting over *sincere* voting was minimal (0.18vs 0.19) In contrast, under the Burger and Rehnquist Courts, decision quality under seniority voting improved markedly, coinciding with increases in both judicial expertise and social learning. Specifically, the probability of mistakes under seniority voting dropped to 0.09, compared to

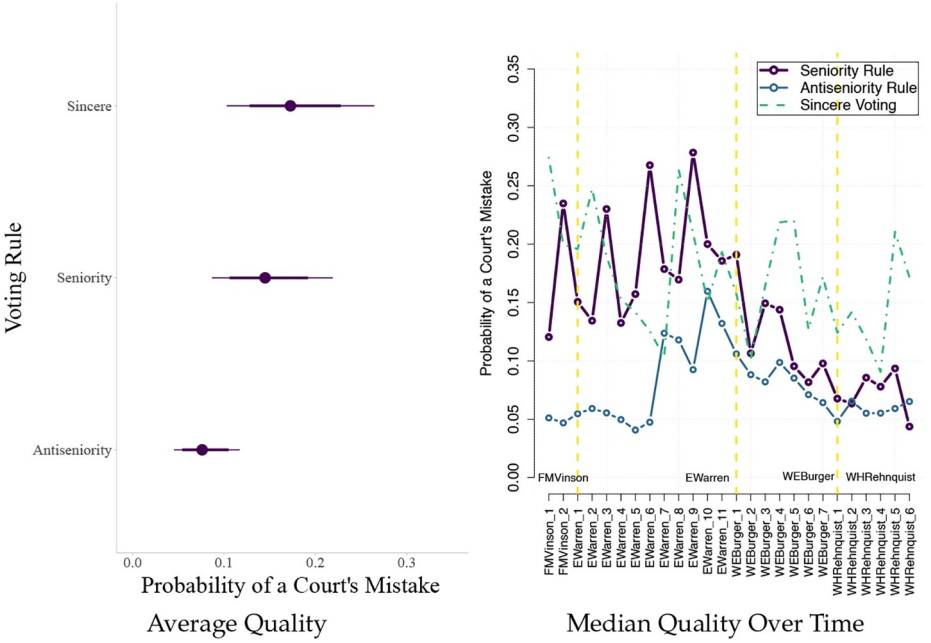


Figure 8. The quality of information aggregation in the court

Note: Each point in the left-hand panel depicts the posterior median of $\hat{Pr}[v_t \neq \omega_t]$ by voting mechanism averaging across both justices and cases, while thick (thin) lines represent 90% (95%) credible intervals. The right-hand panel plots the posterior median of $\hat{Pr}[v_t \neq \omega_t]$ by voting mechanism and natural Court averaging across both justices and cases

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0.16 under *sincere* voting. This improvement also narrowed the gap with the *anti-seniority* rule, reducing the difference in the probability of mistakes from 0.12 to 0.03 [31].

The superiority of the *anti-seniority* mechanism is driven by the higher expertise (relative to bias) of junior justices in the data. Under the seniority voting norm, this higher expertise-to-bias ratio among less experienced justices limits the amount of information revealed through the voting sequence. Figure 9 shows that while justices' biases are largely unrelated to the voting order, information quality significantly improves for those voting later. The main exception is the first voting position, reserved for the Chief Justice regardless of their tenure. Given these patterns, allowing junior justices to vote first under an *anti-seniority* rule amplifies the influence of more expert justices through social learning, thereby improving the quality of the Court's dispositions relative to seniority voting [32].

6. Extensions and concluding remarks

In this paper, we present evidence of social learning in the Supreme Court, whereby junior justices incorporate relevant information about case facts from the votes of their senior colleagues. We develop and estimate a model of sequential voting to quantify this learning process under the seniority norm and assess its impact on decision-making relative to alternative voting rules. Given the non-experimental nature of the institutional setting, this analysis would not be feasible using a reduced-form approach alone.

Consequential model − Our baseline model, like conventional ideal point models, assumes that justices have *sincere* preferences – i.e. they care only about their individual vote. This allows us to isolate the informational effects of social learning from preferences and private information, abstracting from equilibrium behavior. In Supplementary Online Appendix G, we extend the model to allow for *consequential* preferences, where justices care

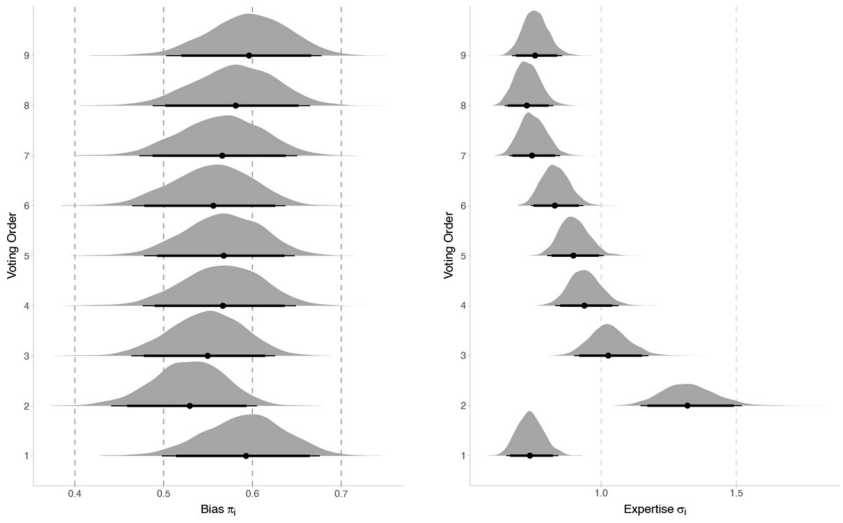


Figure 9. Preferences, expertise and seniority voting

Note: Both panels show the posterior distribution of bias and expertise for each voting position, with thick (thin) lines representing 90% (95%) credible intervals

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about the Court's final disposition, reached by majority rule. This extension captures both learning from senior votes and the strategic influence exerted on junior justices. Using this framework, we compare sequential and simultaneous voting mechanisms in a setting where all justices act strategically [33].

We find that incorporating pivotality considerations improves decision-making under both sequential and simultaneous voting relative to *sincere* voting. Among strategic environments, seniority-based sequential voting slightly outperforms simultaneous voting in aggregating information.

Conformity model − Beyond the reduced-form evidence in favor of social learning relative to conformity incentives presented in Section 2.2, we write down and structurally estimate a *conformity* model in Supplementary Online Appendix H. In this alternative account justices care about the appropriateness of their decision based solely on their private information, as in the *sincere* voting model, but also derive direct utility from aligning with the majority, as in Callander (2008). Under this framework, justices attend to the votes of their peers not to learn about the consequences of their decision, as in our *sequential* model, but rather to assess the likelihood that their own vote coincides with the ultimately prevailing alternative.

We compare the explanatory power of the *conformity* model relative to the *sequential* and *sincere* models. First, we estimate a positive and statistically significant conformity bonus, rejecting the *sincere* model in favor of the *conformity* model. We then compare model fit both in- and out-of-sample, accounting for variation in observed votes and differences in model complexity. In both exercises, the *sequential* and *conformity* models fit the data significantly better than the *sincere* framework, underscoring the relevance of incorporating previous choices in justices' decisions. However, among models that incorporate the voting sequence, the *sequential* model significantly outperforms the *conformity* model, indicating that Bayesian updating and social learning provide a better explanation of conference voting than a behavioral framework that incorporates the voting history solely to conform with the majority.

Concluding remarks − Although we quantify social learning in conference and evaluate the decision-quality implications of the seniority voting norm, conference votes represent only one component of a justice's broader decision-making process. Before reaching the merits, justices must also decide which cases to hear. We leave for future work a systematic assessment of justices' informational constraints and the extent of learning in agenda setting, namely, how justices acquire and aggregate information when selecting cases from the pool of petitions and deciding which to grant or deny.

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Notes

[1.] We rely on the conference votes obtained from the private papers of justices Brennan, Douglas, Powell and Blackmun.

[2.] For any given case, we define the quality of the Court's disposition at conference as the probability that the Court disposes the case consistent with the underlying body of the law.

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Under this metric, the highest quality is achieved by a perfectly informed justice who knows precisely how the law applies to a given case.

[3.] In our data, at least one justice changed their positions in 42% of cases from the initial conference vote on the merits to the official report vote stated in Court's opinions, while individual justices switched their votes on average 11% of the time.

[4.] The Chief Justice is considered the most senior member if they are in the majority coalition.

[5.] Our main results remain unchanged even if we discard all cases decided before the start of the Burger Court (i.e. before June 1969).

[6.] For cases involving multiple rounds of conference voting on the merits, we rely on the first and most preliminary round of votes.

[7.] If a formally decided case contains more than one round of cert votes, we take the last round of votes to capture the decision-making at the selection stage.

[8.] Table A.4 in the Appendix lists all the controls included for estimation.

[9.] In particular, these fixed effects account for seniors' characteristics that remain fixed per natural court and affect a junior justice's disposition, irrespective of seniors' voting position. In addition, they control for court characteristics that remain stable across cases decided within a natural court, such as the composition of the Court's clerks, which vary as departing justices leave the bench and new justices arrive.

[10.] For instance, in uncontroversial cases where all justices have precise information about the correct implementation of the law, justices would vote unanimously on the merits with perfect serial correlation, even though no learning is taking place.

[11.] This approach relies on the randomization inference mechanism of Efron and Tibshirani (1994).

[12.] We classify justices as conservative if their Segal-Cover score falls between 0 and 0.5, and as liberal if the score falls between 0.5 and 1.

[13.] This figure is based on estimates from equation (1), as reported in column (2) of Appendix Table A.4, which interacts the effect of senior justices' votes by their ideology and qualifications at the time of nomination using Segal-Cover scores.

[14.] By contrast, Spenkuch *et al.* (2018) report strong serial correlation in roll-call votes among US House representatives under alphabetical voting procedures but do not find significant learning effects when comparing observed vote patterns to a similar *null distribution*.

[15.] Figure A.3 confirms that case characteristics are statistically indistinguishable before and after departures.

[16.] Callander (2007) and Callander (2008) show that incorporating a behavioral desire to conform into voting models with private information can preserve, and in some cases enhance, information aggregation.

[17.] On average, a justice passed and voted last in 1.3% of the cases they participated in, with substantial variation by seniority (see Figure I.1 in the Appendix).

[18.] Figure A.5 in the Appendix shows that these differences are not only statistically indistinguishable from zero but also comparable to those obtained under random permutations of voting order.

[19.] See Section B.2 in the Appendix, where we simulate the estimates from the test for discontinuous changes in vote dependence for simulated data generated under the *sequential voting* model, considering both homogeneous and heterogeneous voters.

[20.] Appendix G extends the model to consider justices with *consequential* preferences about the outcome of the Court in conference.

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[21.] Iaryczower and Katz (2015) establish the connection between the *spatial voting* model and the model with information in the context of judicial behavior.

[22.] In the baseline empirical model, we directly estimate π_{it} and subsequently recover justices' ideal points z_i in a second step. Appendix E.1 details this procedure and demonstrates that similar results are obtained when we estimate a model that recovers ideal points and alternative-specific parameters directly.

[23.] A predicted vote in the model, $\widetilde{v}_{it}$ is obtained from the estimated voting probability $\widetilde{\gamma}_{it}$, as $\widetilde{v}_{it} = \mathbf{1}_{\{\widetilde{\gamma}_{it} \geq 0.5\}}$. Under the Bayesian framework used for estimation, model fit depends on the posterior distribution of the model parameters and thus reflects uncertainty from posterior sampling, allowing us to report credible intervals around these performance estimates.

[24.] Computing exact leave-one-out cross-validation requires refitting the model once for each observation, which is computationally infeasible in practice. Instead, we approximate LOO using the *Pareto smoothed importance sampling* (PSIS) method of Vehtari *et al.* (2017), which relies on posterior simulations from the original model to efficiently estimate out-of-sample predictive accuracy.

[25.] Appendix F.3 presents the relationship between all covariates and the estimated preference bias parameters.

[26.] In Appendix E.2, we provide two alternative approaches to recover justices' ideal points z_i. First, we estimate the *sequential voting* model using the ideological classification of individual votes. Second, we estimate the *sequential voting* model by directly recovering z_i, β_t and λ_t, imposing standard distributional assumptions on ideal points and case parameters. The results are remarkably similar across specifications.

[27.] In particular, justices' preferences are consistent with their ideological Segal-Cover scores, their party affiliation and the party of their nominating president (see Figure F.5 in the Appendix).

[28.] Figure F.6 in the Appendix shows the relationship between justices' expertise and relevant covariates for *Basic Rights* cases.

[29.] The lowest correlation in expertise estimates across issue areas is observed between *Economics* and *Basic Rights* cases (cor = 0.84), while the highest occurs between *Basic Rights* and *Criminal* issues (cor = 0.94).

[30.] Still, as shown in the left panel of Figure F.4, justices' preferences remain a fundamental source of variation. When comparing predicted vote choices under the *sequential voting* model and the ideological *spatial voting* model, both models agree in 75% of cases on average, though with substantial heterogeneity across justices. The right panel of Figure F.4 shows that agreement between models varies with case characteristics.

[31.] Figure F.10 in the Appendix reports results for *Criminal* and *Economics* cases. In both issue areas, *anti-seniority* voting yields the highest quality. For *Criminal* cases, *sincere* voting also slightly outperforms seniority voting.

[32.] Figure F.3 in the Appendix shows the relationship between the Bayesian probability ι_i and signal precision. We find no clear association between type and expertise across justices.

[33.] We focus on the longest-standing natural Court in our data: the Burger Court (1975–1980).

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Supplementary material

The supplementary material for this article can be found online.

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